

Basic Walkthrough Reverse Engineer in CTF

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Introduction

This is a walkthrough document we prepared to help you get started with reverse engineering in Capture the Flag (CTF) challenges. It covers key concepts, tools, and techniques commonly used in CTF reverse engineering tasks, and includes step-by-step examples to guide your learning.

Demo example challenge name: Debug

This is a beginner-friendly reverse engineering challenge designed to test basic debugging skills. The program checks for the presence of a debugger using `IsDebuggerPresent()`. If no debugger is detected, the program exits immediately, making it essential to run under a debugger to proceed. Once inside, the user is prompted to enter a flag. The correct flag is hardcoded and validated against the input.



```
Command Prompt
C:\Users\User\Desktop\Jica_CTF_Training\demo>debug.exe
CTF Challenge: Debug me if you can.
No debugger detected. Exiting.
C:\Users\User\Desktop\Jica_CTF_Training\demo>
```

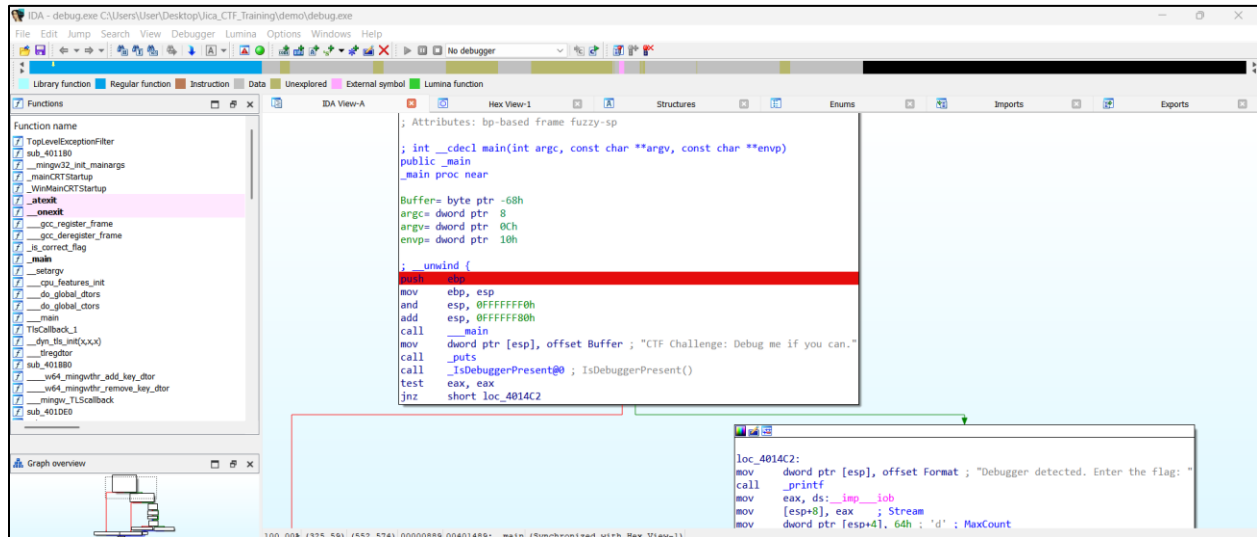
List of tools:

- IDA (<https://hex-rays.com/ida-free>) or (<https://hex-rays.com/ida-pro>)
- Ghidra (<https://github.com/NationalSecurityAgency/ghidra>)
- X64dbg/x86dbg (<https://x64dbg.com/>)

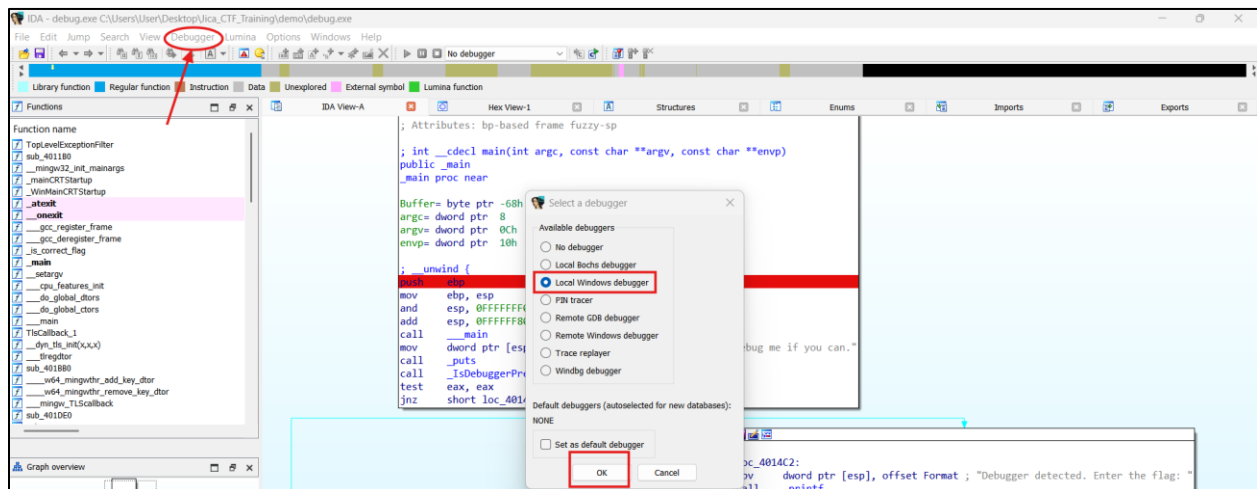
Recommendation: All these tools should only be used in a sandboxed environment such as a virtual machine designed for analyzing unknown binary, do not attempt to analyze unknown binary on your host operating system.

For basic dynamic analysis (using IDA)

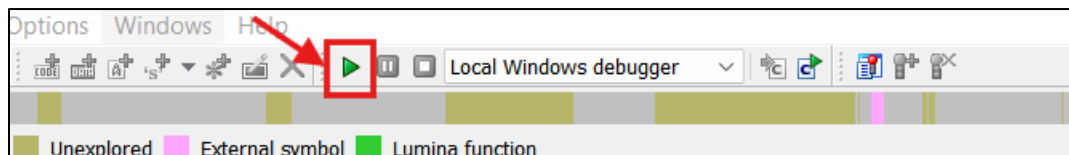
Step 1 load or drop the file into IDA and add the breakpoint by right clicking on the code > selecting Add Break Point (F2 key)

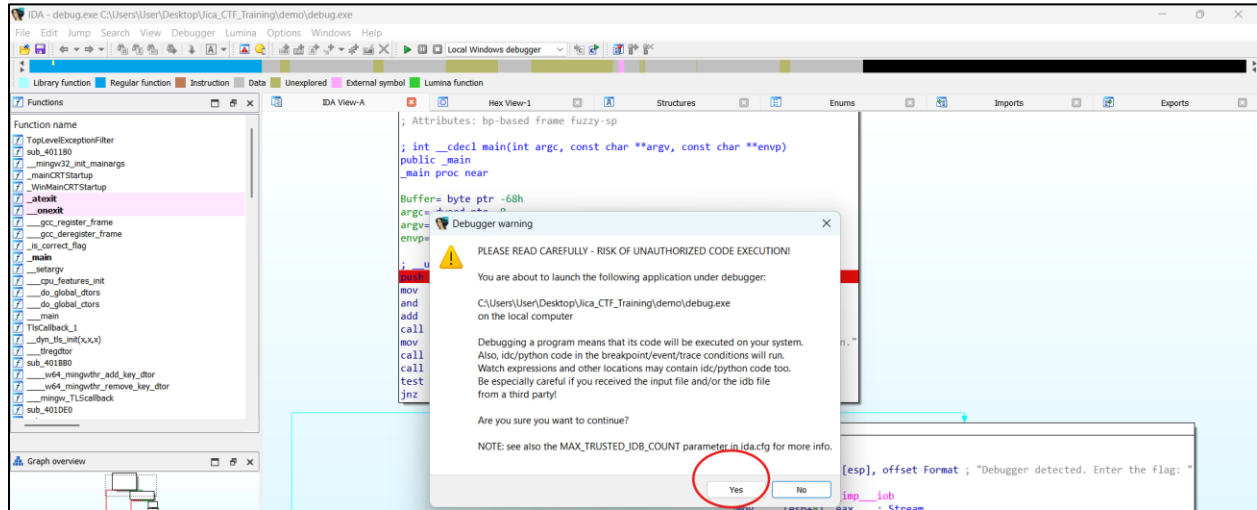


Step 2: Then click on Debugger as the top and select the debugger.



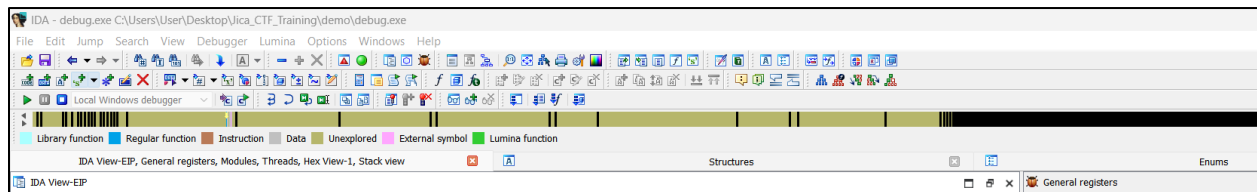
Step 3: Please help click on the play button at the top to run the debugger.



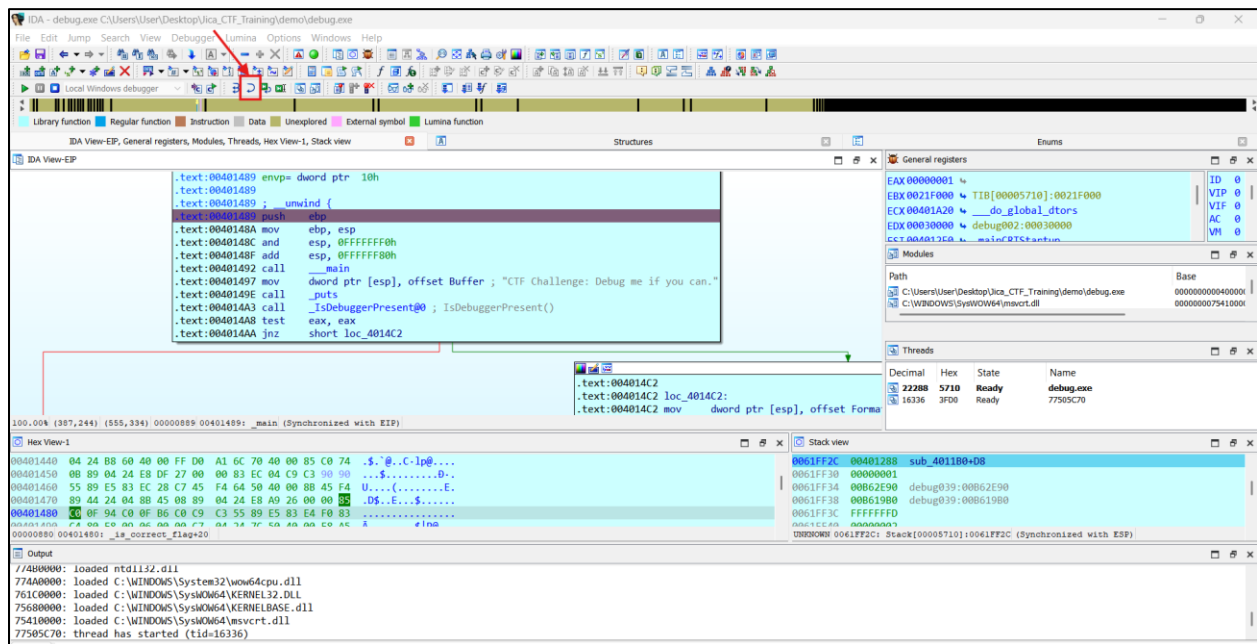


Please click Yes.

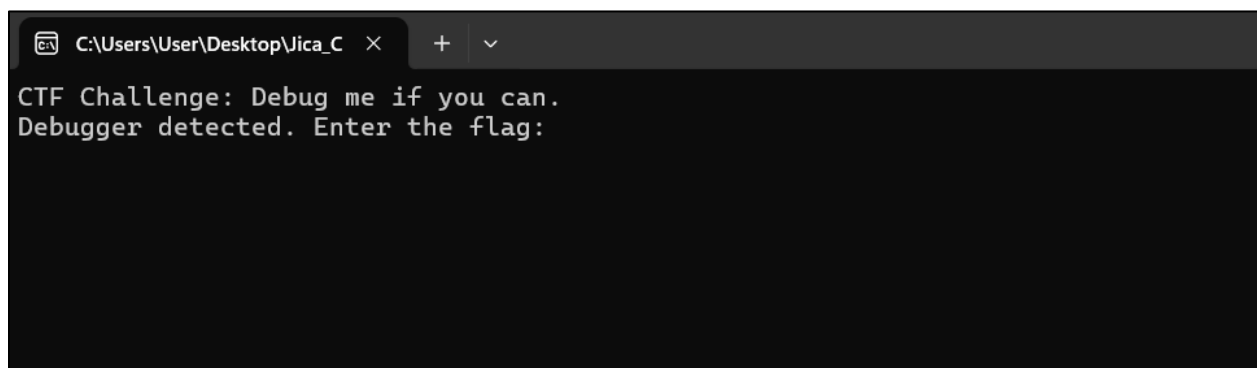
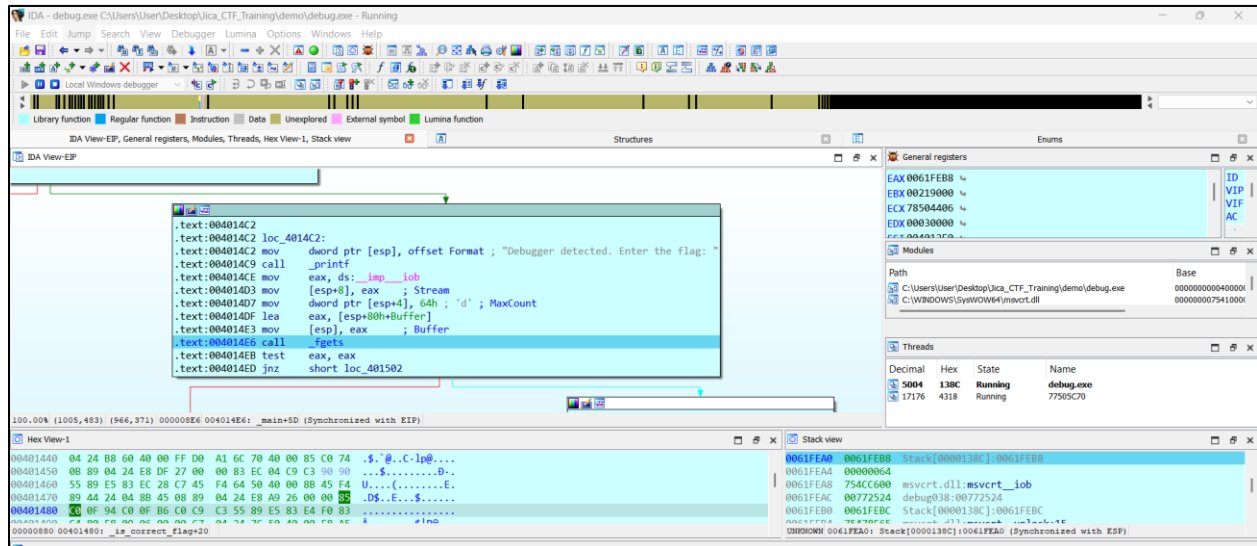
Step 4: go to View > Toolbars > show all



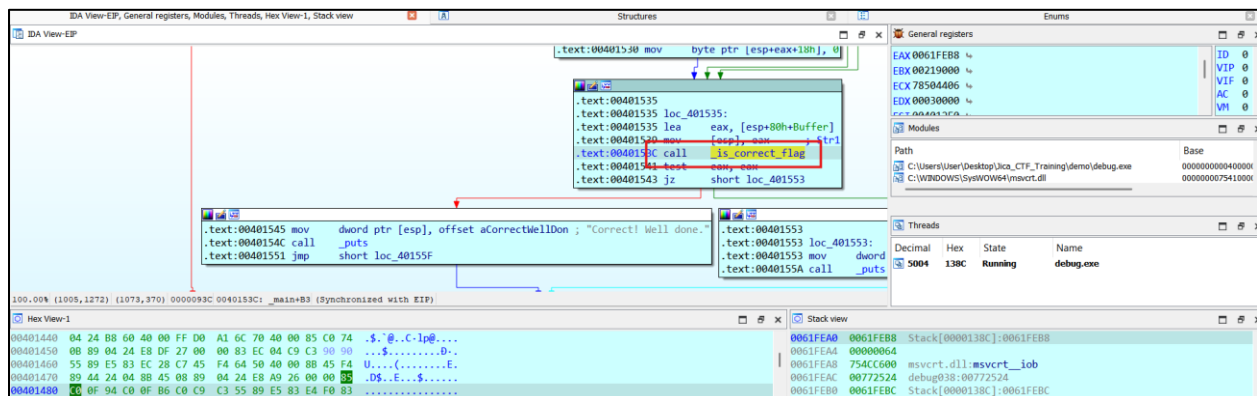
Step 5: We click on step over button to move the code that we want to stop

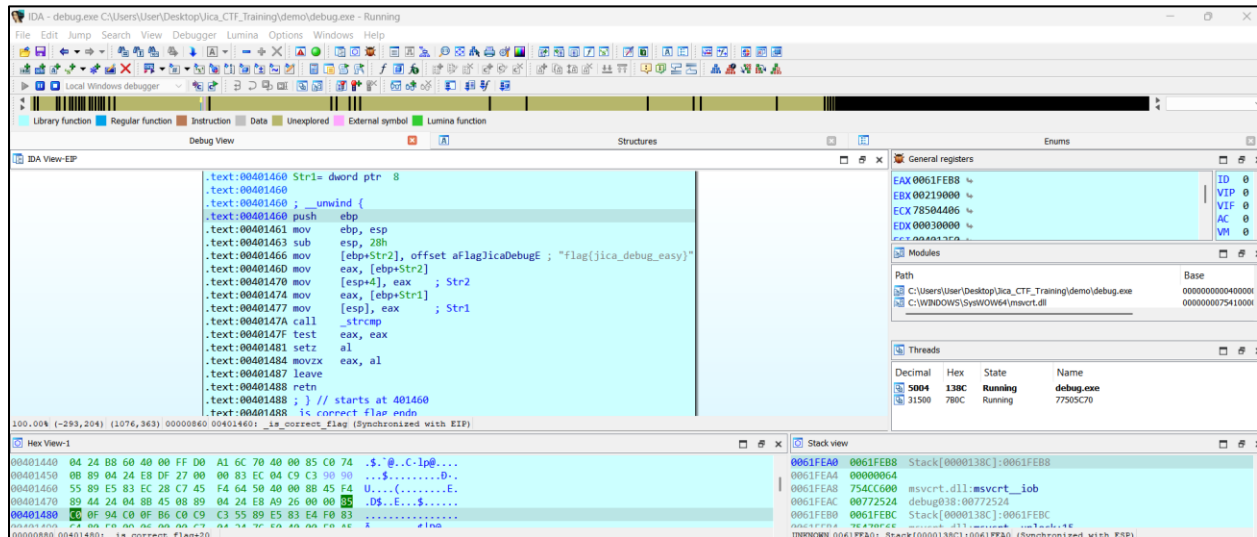


Step 6: After we move the code it stop here and we also looking the program screen.

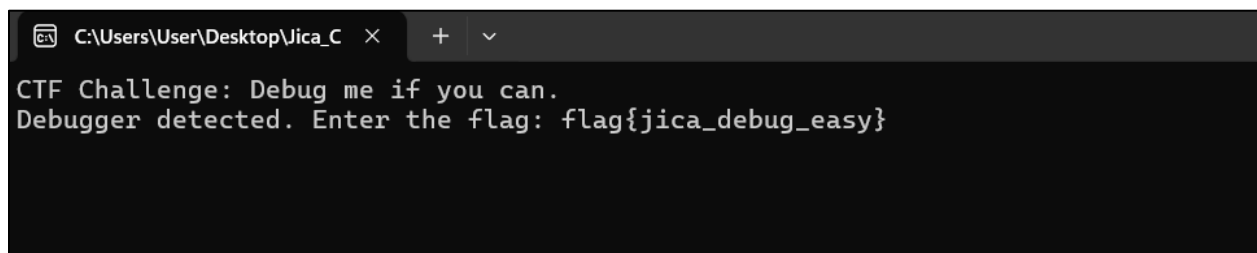


Step 7: in this case we need to look for the flag pattern. Scroll down to find the function named “_is_corrent_flag” and click on it.

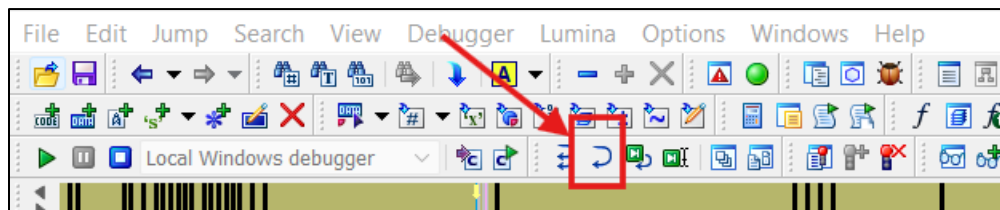


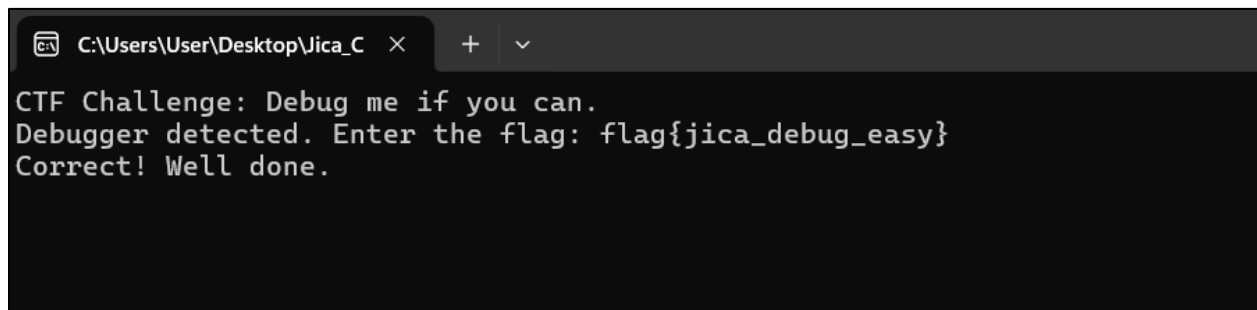
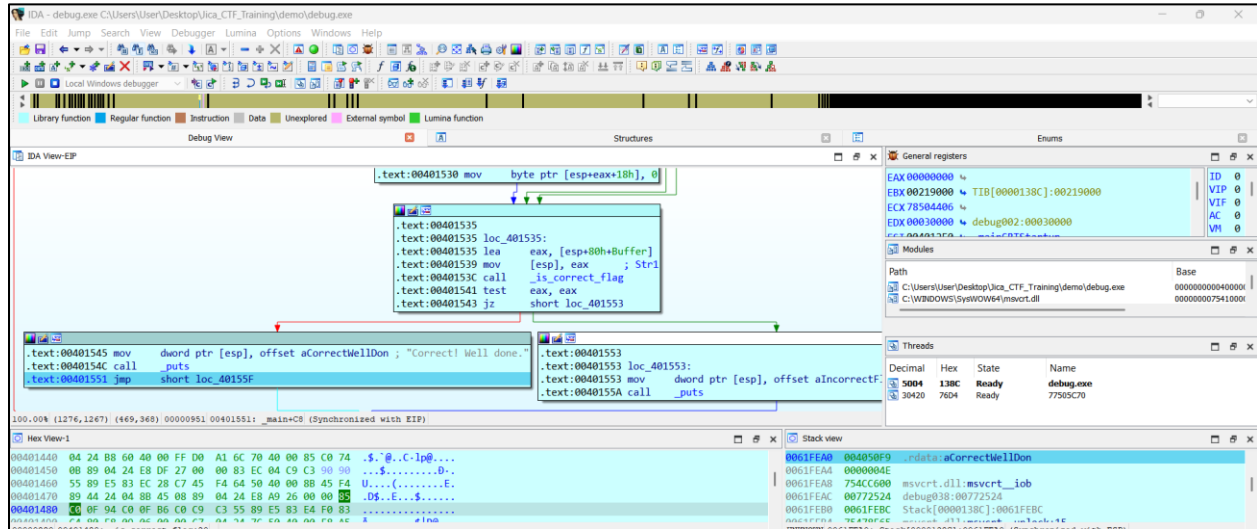


Step 8: After we click on the function we will see the flag. So, we can copy and paste to input to verify the result.



Step 9: After the input, the debugger will start to move to the next instruction. For this we just need to move forward using the step over button like before.





Step 10: Move until passed the message “Correct! Well done” and the flag is **flag{jica_debug_easy}**